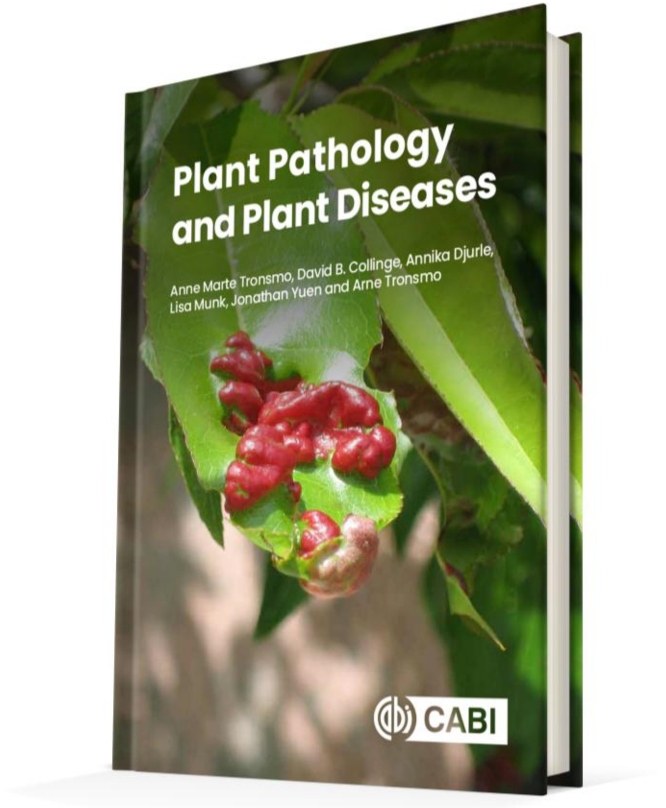

Plant Pathology and Plant Diseases

TEACHING MATERIALS

Fig. 15.1 Ergot caused by *Claviceps purpurea*, here in hybrid rye. (P. Waern, Courtesy of Swedish Board of Agriculture.)

Fig. 15.2 Diseases in barley caused by seedborne pathogens. (A) Bipolaris root and foot rot caused by *Bipolaris sorokiniana*. (B) Net blotch caused by *Pyrenophora teres*. (P. Waern, Courtesy of Swedish Board of Agriculture.)

Fig. 15.3 (A) Fireblight (Erwinia amylovora) on pear (B) Incineration of trees with fireblight. (A,B; © Department of Plant Protection Biology, SLU, Alnarp.)

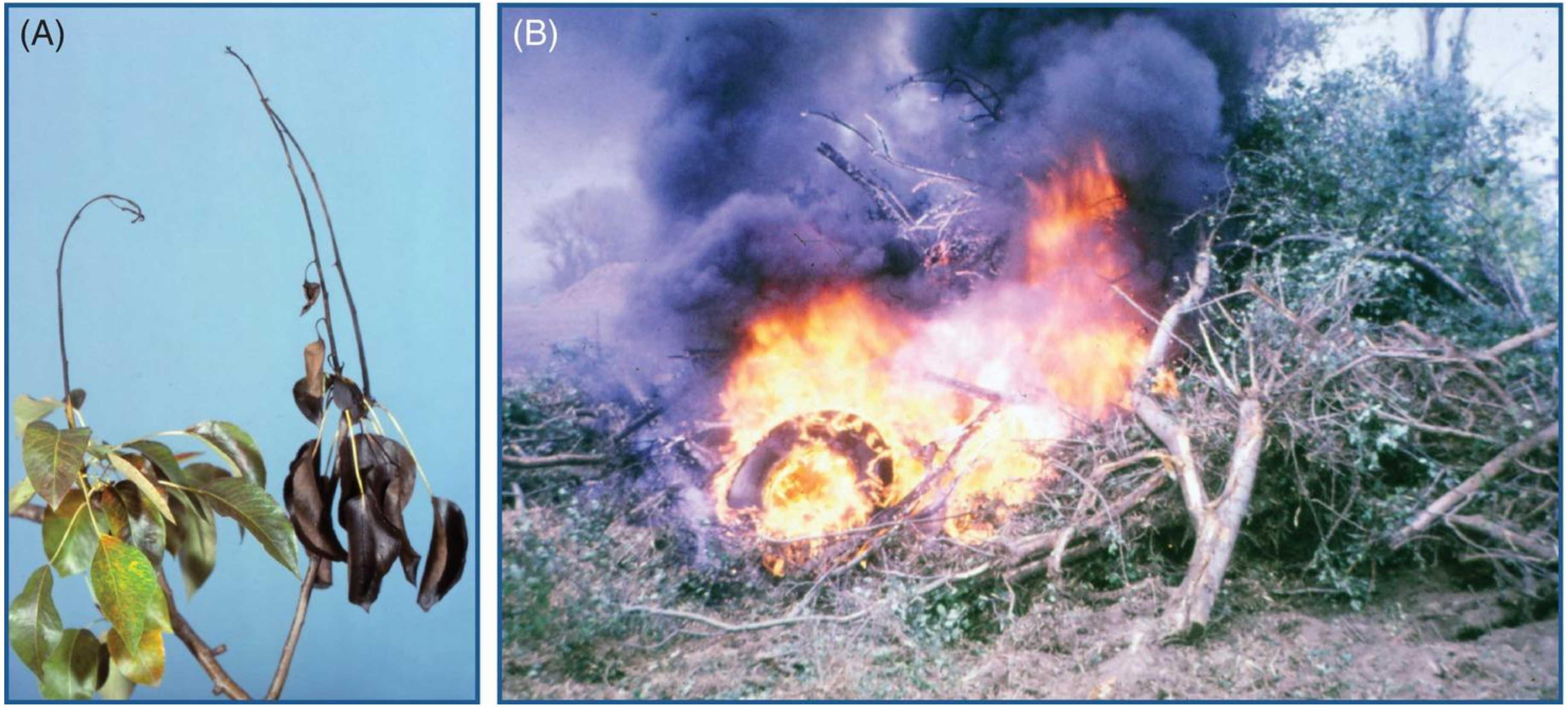

Fig. 15.4 (A) An elm tree slowly dying from Dutch elm disease. (B) Removal of dead elm trees. (© J. Stenlid.)

Fig. 15.5 Viral diseases in potato can cause huge yield losses. It is very important to remove infected plants in the seed potato fields. (© D-R. Blystad.)

Fig. 15.6 Chocolate spot caused by *Botrytis fabae* in faba bean (*Vicia faba*). (©A. Djurle.)

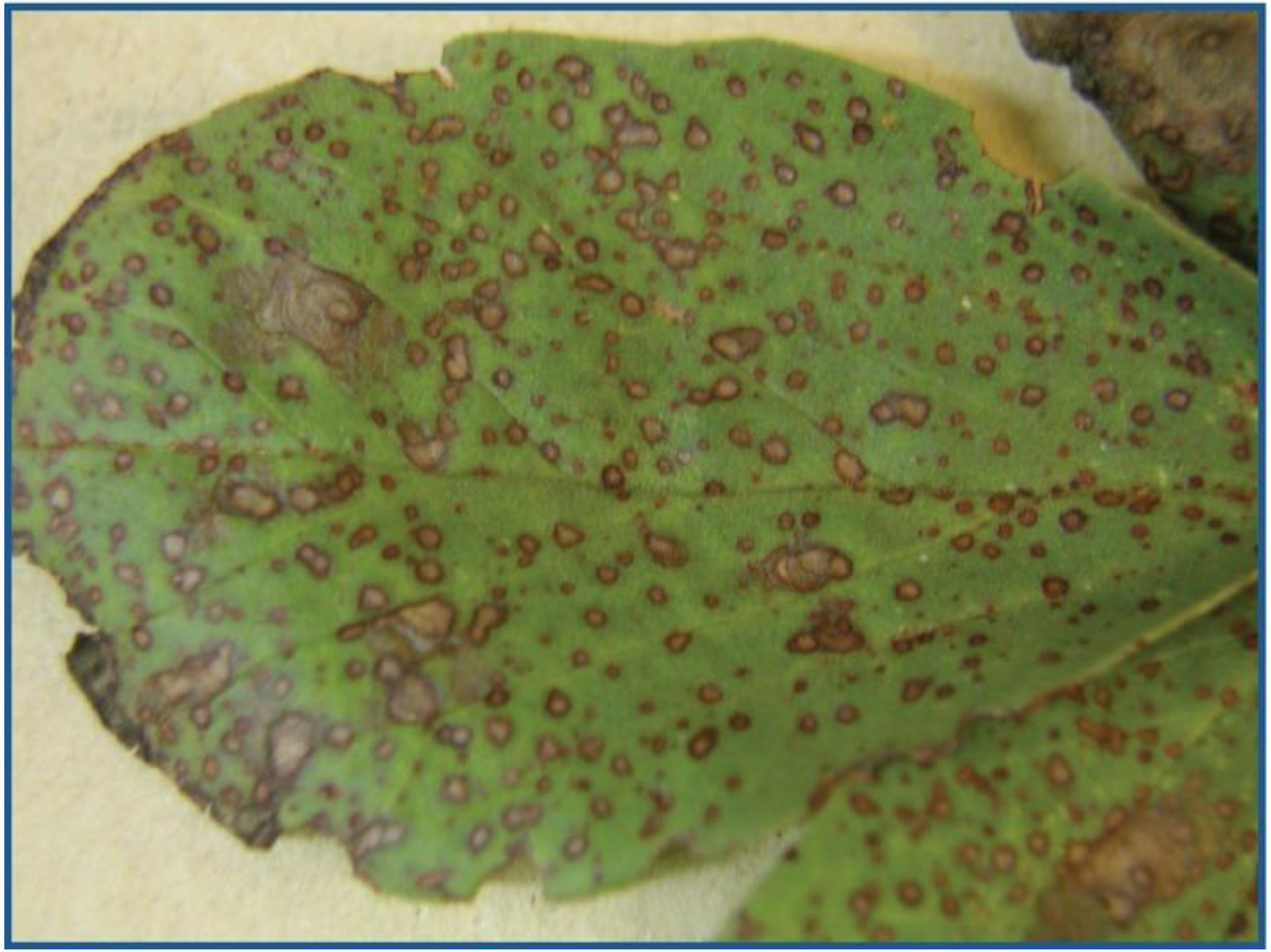

Fig. 15.7 Clusters of aecia on wild barberry (*Berberis vulgaris*), the alternate host of stem rust (*Puccinia graminis*) and stripe rust (*P. striiformis*). (A) Aecia on a shoot. (B) Aecial cups on the underside of a leaf. (©A. Djurle.)

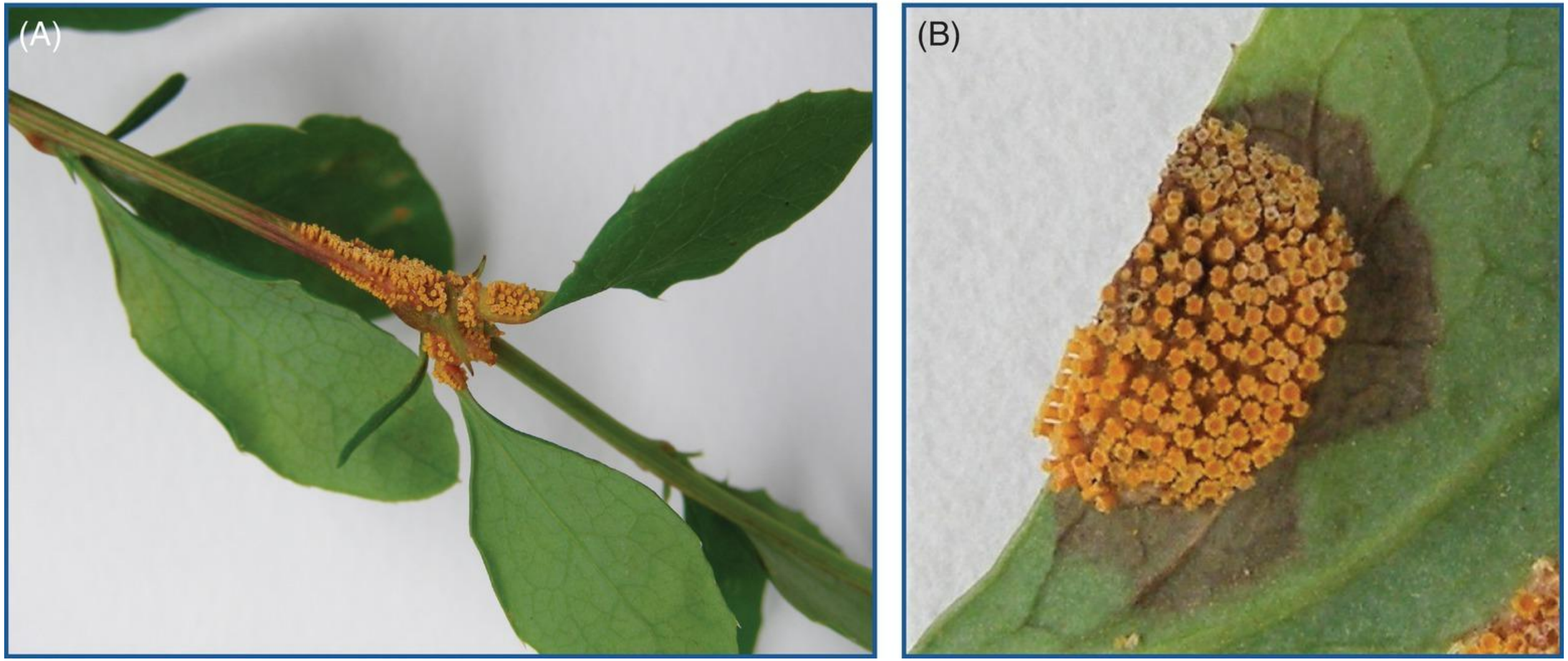

Fig. 15.8 Cucumber green mottle mosaic virus is very infectious. Infected plants should be recognized and removed at an early stage in the culture. (© D-R. Blystad.)

Fig. 15.9 Fruit tree with canker (*Neonectria ditissima*). (© SLU Alnarp.)

Fig. 15.10 Ramularia leaf spot in barley (*Ramularia collo-cygni*). (A) Slightly rhomboid 1–2 mm spots. (B) Ramularia leaf spot, net blotch and powdery mildew on the same leaf. (©A. Djurle.)

Fig. 15.11 Ploughing and other deep tillage methods have both 'pros' and 'cons'. Buried crop residues, and pathogens on them, will decompose faster than on the ground surface. (© L. Munk.)

Fig. 15.12 (A) Winter wheat plants with tan spot (caused by *Pyrenophora tritici-repentis*) and straw residues with pseudothecia from the previous year's winter wheat crop. (B) Eyespot in winter wheat (caused by *Oculimacula yallundae*). (© Swedish Board of Agriculture, P. Waern.)

Fig. 15.13 A severe case of common scab caused by *Streptomyces scabiei*.
(© Department of Plant Protection Biology, SLU, Alnarp.)

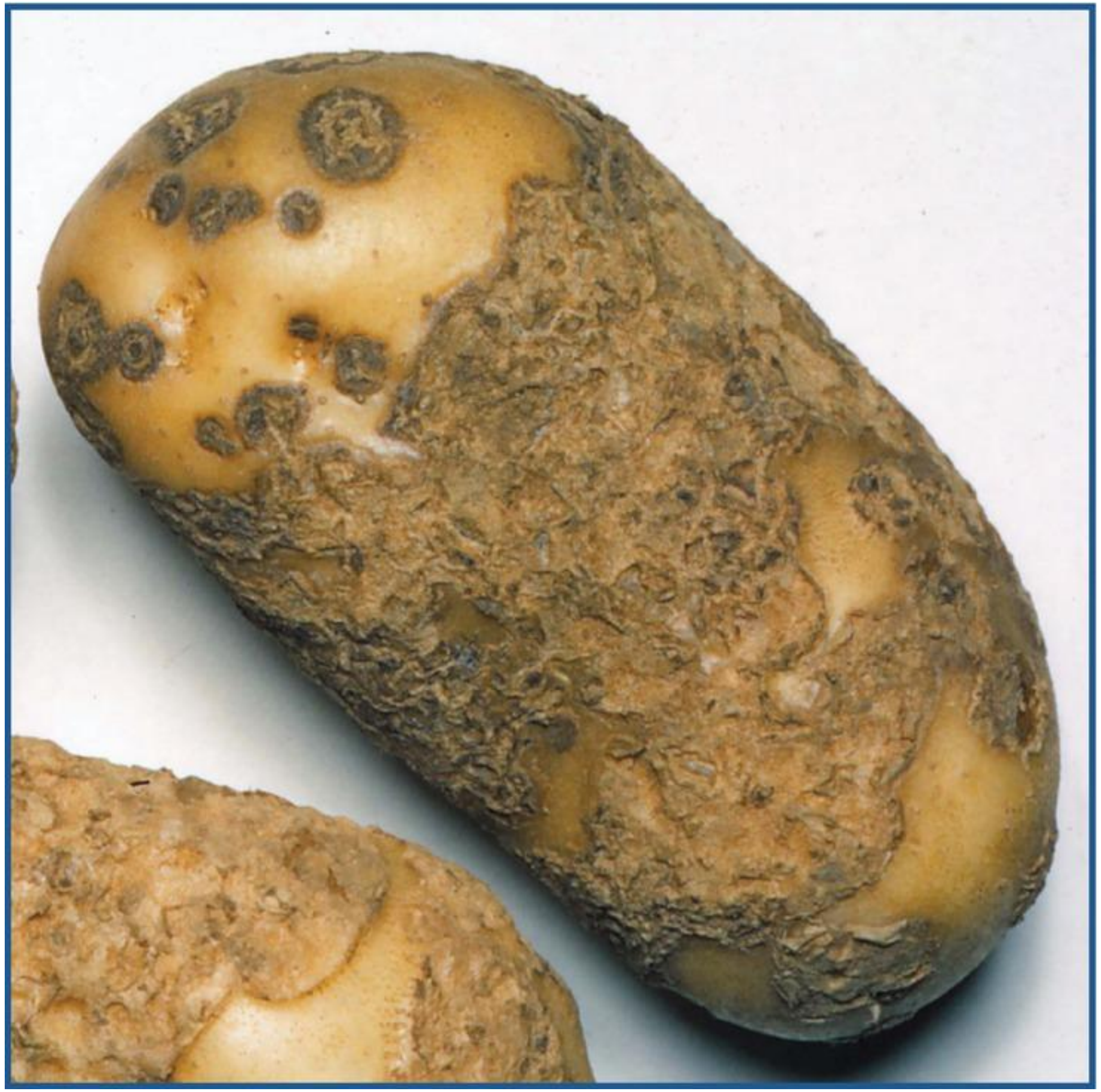

Fig. 15.14 Plants infected with wheat dwarf virus (WDV). An effect from early sowing of winter wheat when vectors were active. (A. Djurberg, Courtesy of Swedish Board of Agriculture.)

Fig. 15.15 Spring oilseed rape with black spot (*Alternaria brassicae*). (©A. Djurle.)

Fig. 15.16 Shallow sowing increases the risk for up-freezing of autumn-sown cereals. ***Cephalosporium stripe*** **is caused by** ***Hymenula cerealis*** **that grows in the vascular system of the host plant.** (©A. Djurle.)

Fig. 15.17 Diseases caused by quarantine pathogens. (A) Ring rot in potato caused by *Clavibacter sepedonicus*. (B) Brown rot caused by *Ralstonia solanacearum.* (© Department of Plant Protection Biology, SLU, Alnarp.)

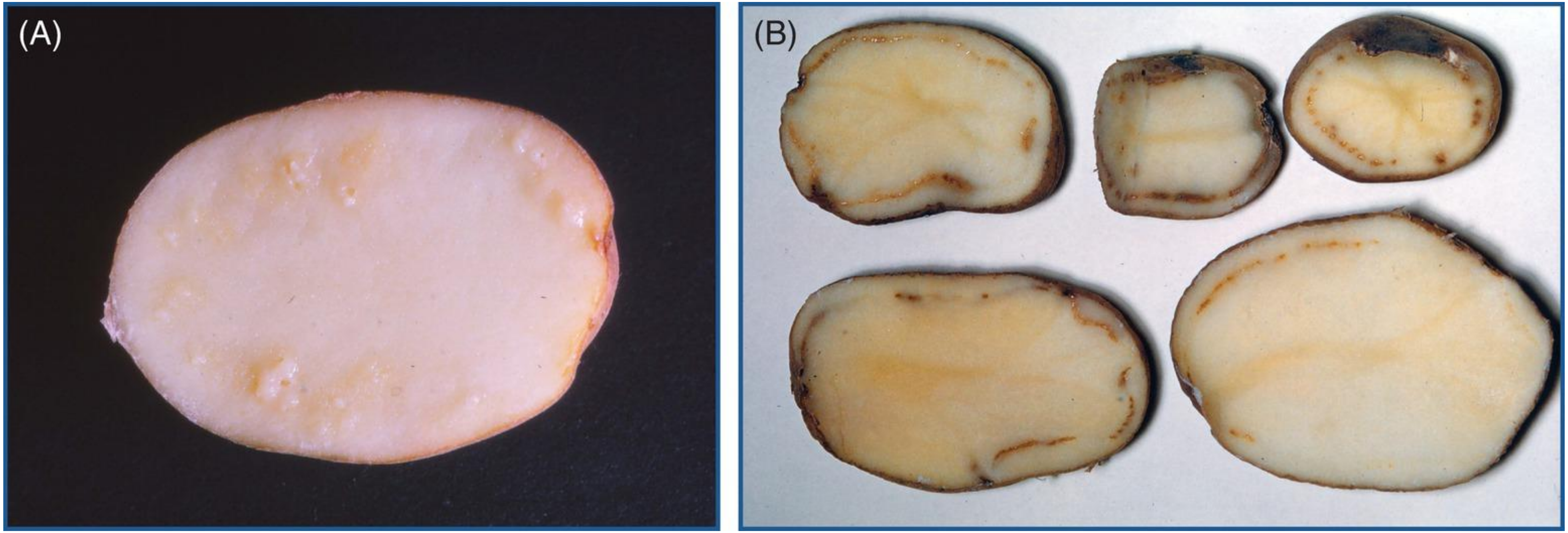

Fig. 15.18 Plum pox virus showing typical ringspot symptoms in plum (*Prunus domestica*). This virus directly affects fruit quality and yield and is considered as a quarantine virus in several countries. (© D-R. Blystad.)